Arjuna JEE 2025

Mathematics

Permutations and Combinations

DPP: 9

Q1 The number of divisors of 9600 including 1 and 9600 are
(A) 60
(B) 58
(C) 48
(D) 40

Q2 The sum of the divisors of $2^5 . 3^7 . 5^3 . 7^2$, is
(A) $2^6 . 3^8 . 5^4 . 7^3$
(B) $2^6 . 3^8 . 5^4 . 7^3 - 2.3.5.7$
(C) $2^6 . 3^8 . 5^4 . 7^3 - 1$
(D) none of these

Q3 Number of divisors of the number $\mathrm{N} = 2^3 \cdot 3^5 \cdot 5^7 . 7^9$ which are perfect square, is
(A) 24
(B) 60
(C) 119
(D) 120

Q4 The sum of all positive divisors of 960 is
(A) 3048
(B) 3087
(C) 3047
(D) 2180

Q5 The number of divisors of $a^p b^q c^r d^s$, where a, b, c, d are primes and $p, q, r, s \in \mathrm{N}$ excluding 1 and the number itself, is
(A) $pqrs$
(B) $(p+1)\ (q+1)(r+1(s+1)\ -\ 4$
(C) $pqrs\ -2$
(D) $(p+1)\ (q+1)(r+1(s+1)\ -\ 2$

Q6 If $\mathrm{N} = 2^3 \cdot 3^4 \cdot 5^2 \cdot 7^1$ then product of all divisors of N which are divisible by 5 is
(A) N^{20}
(B) N^{40}
(C) $(5\ \mathrm{N})^{20}$
(D) $(5\ \mathrm{N})^{40}$

Q7 If total number of even divisors of 1323000 which are divisible by 105 is $2^{\mathrm{k}} - 10$, then k is equal to
(A) 5
(B) 6
(C) 7
(D) 8

Q8 The number of divisors of the form $4\mathrm{k} + 2, \mathrm{k} \geq 0$ of the integer 240 is
(A) 4
(B) 8
(C) 10
(D) 3

Q9 Let N be the number of points (x, y, z) in space such that $x + y + z = 12$, where $x, y, z \in \mathrm{N}$. The number of divisors of N is equal to
(A) 2
(B) 4
(C) 9
(D) 16

Q10 If $\mathrm{p}, \mathrm{q}, \mathrm{r}$ are prime numbers α, β, γ are positive integers such that L.C.M. of α, β, γ is $\mathrm{p}^3 \mathrm{q}^2 r$ and greatest common divisor of α, β, γ is pqr then the number of possible triplets (α, β, γ) will be
(A) 6
(B) 12
(C) 72
(D) 96

Q11 The total number of ways in which 30 mangoes can be distributed among 5 persons is
(A) ${}^{30}\mathrm{C}_5$
(B) 30^5
(C) 5^{30}
(D) ${}^{34}\mathrm{C}_4$

Q12 The number of ways of distributing 5 identical balls into three boxes so that no box is empty

(each box being large enough to accommodate all balls), is
(A) 3^5
(B) 5^3
(C) 15
(D) 6

Q13 Four boys picked up 30 mangoes. In how many ways can they divide them, if all mangoes be identical?
(A) 5400
(B) 5448
(C) 5456
(D) 5478

Q14 The number of ways in which 10 identical apples can be distributed among 6 children so that each child receives atleast one apple is:
(A) 126
(B) 252
(C) 378
(D) None of these

Q15 In how many ways can 15 identical blankets be distributed among six beggars such that everyone gets atleast one blanket and two particular beggars get equal blankets and another three particular beggars get equal blankets.
(A) 10
(B) 12
(C) 15
(D) 18

Q16 The number of integral solutions of $x+y+z=0$ with $x, y, z \geq -5$
(A) 120
(B) 136
(C) 144
(D) None of these

Q17 The number of positive integral solutions of $15<x_1+x_2+x_3 \leq 20$, is equal to
(A) 785
(B) 685
(C) 1150
(D) None of these

Q18 The number of positive integral solutions of the inequality $x+y+z \leq 20$ is
(A) 1008
(B) 1028
(C) 1108
(D) 1140

Q19 The number of non-negative integral solutions of $x_1+x_2+x_3+x_4 \leq n$ (where n is a positive integer) is:
(A) ${}^{n+3}C_3$
(B) ${}^{n+4}C_4$
(C) ${}^{n+5}C_5$
(D) ${}^{n+4}C_n$

Q20 In how many ways the number 18900 can be split in two factors which are relative prime (or coprime)?
(A) 6
(B) 7
(C) 8
(D) None of these

Answer Key

Q1	(C)
Q2	(D)
Q3	(D)
Q4	(A)
Q5	(D)
Q6	(D)
Q7	(B)
Q8	(A)
Q9	(B)
Q10	(C)
Q11	(D)
Q12	(D)
Q13	(C)
Q14	(A)
Q15	(B)
Q16	(B)
Q17	(B)
Q18	(D)
Q19	(B, D)
Q20	(C)